Jiacan Li

Southern University of Science and Technology | lijc2023@mail.sustech.edu.cn
Personal Page | GitHub Repository

Research Interests: Dexterous Manipulation, Robot Learning, 3D Perception, and Autonomous Navigation

Education

Southern University of Science and Technology
Sep 2023 – Present
Shenzhen, China

B.Eng. in Robotics Engineering
Department of Mechanical and Energy Engineering

- **Overall GPA: 3.79**/4.0 (Weighted Average: **91.37**/100); **Major GPA: 3.81**/4.0.
- **Core Courses (scores out of 100):** Robot Operating System (99), Robot Modeling and Control (95), Computer Vision (94), Machine Learning for Engineering (97), Robotic Actuation System (94), Mechanical Design (97), Finite Element Theory and Its Engineering Applications (97).

National University of Singapore
Jun 2026 – Present
Singapore

Visiting Research Student
Department of Electrical and Computer Engineering

Publications & Patents

Manuscript Under Review

[M1] Jingfeng Xiong[+], **Jiacan Li**[+], Yanchen Wu, Kexin Mu, Hanqing Ma, Yaojin Xu, Zhuoyang Ding, Xinying Li, ChuHeng Chen, and Chenglong Fu[*], "Kinematic-Guided Control Enables Generalizable and Real-Time-Verifiable Exoskeleton Control for Real-World Deployment." Manuscript under review at *Science Advances*. [+]co-first authors; [*]corresponding author.

Patents and Patent Applications

[P1] China, patent no. CN121081241B, "Control method, control device, electronic equipment and storage medium." **Granted**. **Inventors:** Chenglong Fu, Jingfeng Xiong, **Jiacan Li**, Chuheng Chen, Yanchen Wu, Xinying Li, Youran Xi, and Yaojin Xu.

[P2] China, patent no. CN121733493B, "Exoskeleton control method, exoskeleton control system, electronic equipment and computer program product." **Granted**. **Inventors:** Chenglong Fu, Xinying Li, Jingfeng Xiong, Xiaohe Yu, Yanchen Wu, and **Jiacan Li**.

[P3] China, publication no. CN120459594A, "Resistance exoskeleton mechanism control method, resistance exoskeleton mechanism and underlying controller." **Published patent application**. **Inventors:** Chenglong Fu, Shiquan Yu, Yuquan Leng, Guangshan Wang, Xinying Li, **Jiacan Li**, Wenbing Zhuang, and Jingfeng Xiong.

Research Experience

Visuotactile Friction Mapping for Dexterous Grasping
Jun 2026 – Present

National University of Singapore | Singapore | Advisor: Xingyu Liu

- Built an NVIDIA Isaac Lab simulation platform integrating an xArm robotic arm, a LEAP Hand, and GelSight Mini fingertip sensors for visuotactile perception and dexterous grasping.
- Developed a UIPC-based finite element pipeline to simulate gel deformation and generate tactile images and marker displacements. Optimized gel meshes to reduce element count by 25.9% while preserving contact-surface resolution.
- Generated tactile sequences through controlled pressing and sliding. Developed a temporal model to reconstruct spatially varying surface friction maps from sequences of depth and marker images.
- Integrating friction maps into reinforcement learning for grasp planning and control under heterogeneous friction.

Unified Exoskeleton Control for Multi-Terrain Assistance [M1] [P1]
May 2025 – Aug 2026

Southern University of Science and Technology | Shenzhen, China | Advisor: Chenglong Fu

- Developed a unified neural network-based approach to guide exoskeleton assistance using wearer kinematics during level-ground, ramp, and stair walking, without explicit inverse dynamics modeling.
- Deployed a lightweight model on embedded hardware for real-time inference.

Gait Estimation for Post-Stroke Rehabilitation [P2]
Jan 2026 – Apr 2026

Southern University of Science and Technology | Shenzhen, China | Advisor: Chenglong Fu

- Developed and deployed a neural network-based framework to estimate affected-limb kinematics in real time from contralateral limb motion for lower-limb exoskeleton-assisted stroke rehabilitation.

Adaptive Resistance Control for Exercise Exoskeletons [P3]
Jan 2025 – Mar 2025

Southern University of Science and Technology | Shenzhen, China | Advisor: Chenglong Fu

- Studied and applied Bayesian optimization in a human-in-the-loop framework to personalize exercise exoskeleton resistance levels for individual training goals.

Selected Projects

SplatSLAM: Monocular 3D Reconstruction — Jun 2026

- Extracted and standardized frames from monocular smartphone videos with FFmpeg. Used MASt3R-SLAM to estimate camera trajectories and generate dense colored point clouds without depth sensors.
- Refined point clouds in CloudCompare using statistical outlier removal, voxel downsampling, and manual masking. Initialized 3D Gaussian Splatting (3DGS) with the refined geometry for interactive novel-view synthesis.

ROS 2-Based Mobile Manipulator with Multimodal HRI — Jan 2026

- Built a ROS 2/Gazebo mobile manipulation system integrating OpenCV target detection, MoveIt 2 grasping, and Nav2 navigation for autonomous pick-and-deliver tasks.
- Implemented gesture and voice interfaces with MediaPipe and Vosk. Coordinated missions through state machines and multithreaded ROS 2 execution to maintain responsive interaction during navigation.

6-DOF Robotic Arm Automated Drawing System — May 2025

- Converted images into drawing trajectories using OpenCV skeletonization and waypoint sequencing. Derived an analytical inverse kinematics solution from D–H parameters, mapped planar paths onto cylindrical surfaces, and validated motion in Simscape Multibody before physical execution.
- Designed a spring-loaded compliant pen holder for passive contact regulation during drawing.

Agricultural Robot (ROBOCON 2024) — Jul 2024

- Designed a custom gripper for rice harvesting.
- Integrated robot hardware using soldered serial-port connections and custom printed circuit boards (PCBs) for communication.

Grants, Awards & Honors

- **Outstanding Student Award** — 2024, 2026
- **Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation** — 2025
  "Climbing Program" Special Funds; CNY 20,000; Leader.
- **China International College Students' Innovation Competition** — 2025
  National Bronze Award and Provincial Gold Award.
- **Second Prize Scholarship** (Top 8%, twice) — 2024, 2025
- **Outstanding Student Leader Award** — 2025
- **ROBOCON China Championship** — 2024
  National First Prize (Operation Challenge); National Second Prize (Theme Contest); two National Third Prizes (Biomimetic Legged Robot).
- **Outstanding Volunteer** (Top 3%) — 2024

Skills & Additional Information

- **Programming:** Python, C++, MATLAB
- **Robotics & Simulation:** ROS 2, MuJoCo, NVIDIA Isaac Lab, Gazebo
- **Machine Learning:** PyTorch
- **Computer Vision & Visualization:** OpenCV, RViz2
- **Embedded Systems:** STM32, Arduino
- **CAD & CAE:** SolidWorks, AutoCAD, Abaqus
- **English Proficiency:** TOEFL iBT 5.5/6 (comparable score: 110/120)
- **Certifications:** AHA Heartsaver®, Certified First Aider (Red Cross Society of China)
- **Interests:** Go (board game), amateur 1-kyu

Leadership & Service

First Aid & Emergency Response

- **SUSTech Red Cross — Director, Organization Department.** Completed 144.5 hours of volunteer service. Organized and taught 5+ first aid and emergency response training sessions. Led 10+ large-scale events and competitions, including carnivals and forums.
- **Shenzhen Nanshan District Emergency Rescue Association — Instructor.** Established certification standards and procedures. Trained first aid instructors and responders.
- **Campus First Aid Responder.** Provided first aid in 20+ emergencies, including seizures and traumatic injuries such as fractures.

Other Campus Leadership

- **Student Representative, Department of Mechanical and Energy Engineering.**
- **Head of Operations, Robotics Club.**